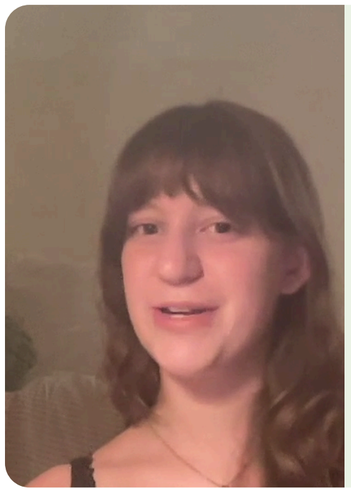
Meet your instructors: undergraduates teaching undergraduates.

UNDERGRADUATE INSTRUCTOR · LESSONS 1 & 2

Chloe King

Unix navigation and Bash scripting



FROM CHLOE'S OWN SESSION

1. **Moves the fridge into the kitchen**

```
HouseGame % mv Decoration/fridge.txt House/Kitchen/
```

2. **Creates a new oven and describes it**

```
Kitchen % touch oven.txt
Kitchen % nano oven.txt
Kitchen % cat oven.txt
A place to bake your cookies!
```

3. **Reads the first wine orders**

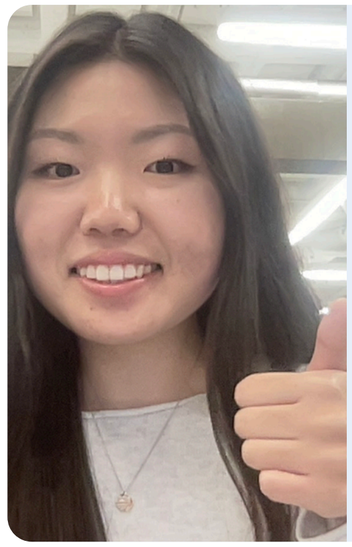
```
WineGame_New % cat orders_round1.csv
```

> "Even something as simple as having two spaces instead of one can completely make the code stop working."
>
> CHLOE, IN HER OWN WORDS

UNDERGRADUATE INSTRUCTOR · LESSON 3

Lindsey Ko

MRI file formats and conversion



FROM LINDSEY'S OWN SESSION

1. **Looks inside a raw scan folder**

```
sub-001 % cd FLAIR_DICOM
FLAIR_DICOM % ls
1.2.840.113619.6.495….-4-1-z43833.dcm   …
```

2. **Converts it with dcm2niix**

```
MRIcron % dcm2niix -f "sub-001_FLAIR_GUI" -p y -z y -o
…/gui_niftis
Found 44 DICOM file(s)
```

3. **Checks what came out**

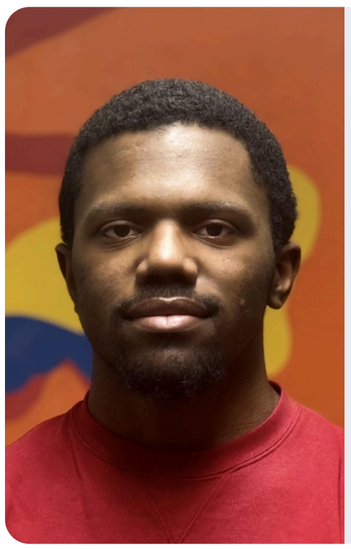
```
ScanConverterGame % ls gui_niftis
sub-001_FLAIR_GUI.json   sub-001_FLAIR_GUI.nii.gz
```

> "Once I had that installed and set up, the dcm2niix started working."
>
> LINDSEY, IN HER OWN WORDS

UNDERGRADUATE INSTRUCTOR · LESSON 4

Osiel Iglesias

Batch processing and remote computing



FROM OSIEL'S OWN SESSION

1. **Looks at the whole dirty load**

```
LaundryGame2 % cd dirty_laundry/round2_big_load
round2_big_load % ls
labcoat_dirty.png  pants_dirty.png  shirt_1_dirty.png  …
```

2. **Logs in to the remote laundromat**

```
LaundryGame2 % bash scripts/fake_ssh.sh
Connecting to ...
Password:
```

3. **Checks the clean results**

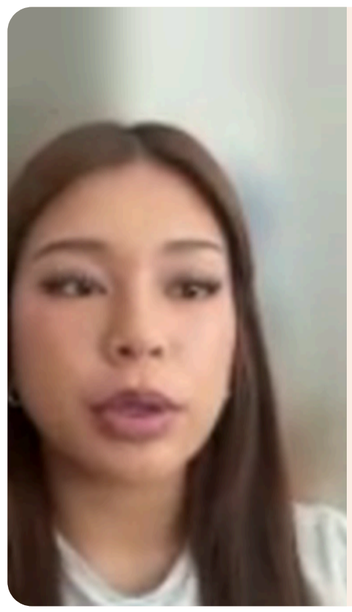
```
clean_laundry % ls
pants_clean.png  shirt_clean.png  socks_clean.png
```

> "…not to memorize every command, it's to understand the overall workflow."
>
> OSIEL, IN HIS OWN WORDS

UNDERGRADUATE INSTRUCTOR · LESSON 5

Tiffany Huang

Brain image review and lesion identification

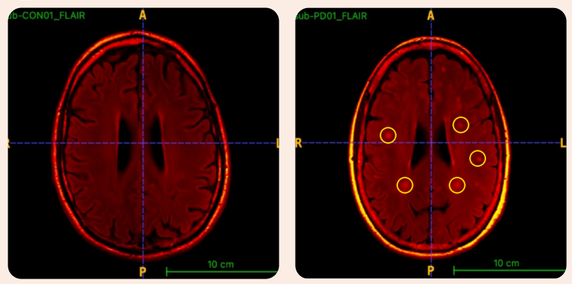


FROM TIFFANY'S OWN SESSION

1. **Opens the lesion lab**

```
~ % cd Desktop/ACLAB/LesionDetectiveGame_CAPSTONE
LesionDetectiveGame_CAPSTONE % ls
converted_niftis  lesion_masks  raw_dicoms  results  …
```

2. **Compares the same slice in ITK-SNAP**



Control — Parkinson's: spots to flag

> "We combine all the skills that we've been learning into neuroimaging."
>
> TIFFANY, IN HER OWN WORDS